PC BOT CAR

Obstacle Avoiding and Obstacle Following Robot Car.

This manual explains how to assemble, wire, upload code, and test a basic robot car using Arduino Uno, IR sensor, and L298N motor driver. It is written for classroom demonstrations, preschool STEM workshops, and beginner robotics activities where the trainer needs a clear step by step build process.

Quick Build Overview

Stage	What to do	Expected result
Mechanical assembly	Fix motors, wheels, battery holder, Arduino Uno, L298N, and IR sensor on chassis.	Car body is ready and stable.
Motor wiring	Connect motors to OUT1 OUT2 and OUT3 OUT4 on L298N.	Both motors can be controlled by driver.
Upload and test	Upload code and test motor direction, sensor response, and movement.	Robot avoids or follows obstacle correctly.

Components Required

Component	Quantity	Purpose
Arduino Uno	1	Main controller for reading sensor and controlling motor driver
L298N motor driver	1	Controls two DC motors
IR sensor module	1 or 2	Detects obstacle or target object
BO motors with wheels	2	Robot movement
Robot chassis	1	Base plate for all parts
Battery pack	1	Power source for motors
Jumper wires	As required	Connections between modules
Screws or double tape	As required	Fixing components on chassis

Safety Before Starting

- Do not connect motor battery directly to Arduino 5V pin.
- Arduino GND and L298N GND must be common, otherwise signals may not work.
- Keep ENA and ENB jumper caps on L298N if speed control wires are not used.
- If L298N becomes very hot, stop the battery supply and check motor wiring.
- Use a proper motor battery pack. A small rectangular 9V battery is usually weak for motors.

Car Assembly Steps

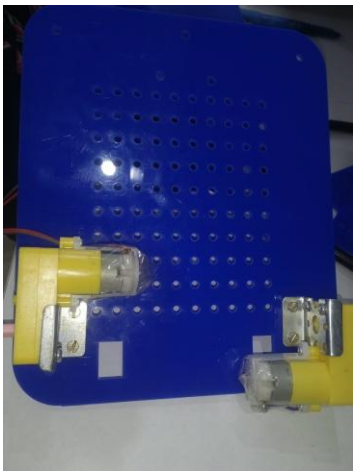
Assembly photos from the actual car build.



Base plate with BO motors fitted on both sides



Caster wheel mounted on the front side of the chassis



Motor position checked before tightening the brackets



Motor bracket fixed with screws on the chassis plate



Side view of BO motor mounted on the chassis



Final Look

Step 1 Mount the Motors

Material: Robot chassis, two BO motors, screws or double tape.

Action: Fix one motor on the left side and one motor on the right side of the chassis. Keep the shaft facing outward so wheels can be attached.

Check: Both motors should be tight and parallel. Loose motors will make the car move in a curved path.

Step 2 Attach the Wheels

Material: Two wheels and motor shafts.

Action: Press the wheels gently onto the motor shafts. Do not bend the motor shaft while pressing.

Check: Rotate both wheels by hand. They should move freely without touching the chassis.

Step 3 Mount Arduino Uno and L298N

Material: Arduino Uno, L298N motor driver, screws or double tape.

Action: Place Arduino Uno near the center and L298N near the motors. Keep enough space for wire connections.

Check: USB cable should be easy to connect for code upload. L298N screw terminals should be easy to access.

Step 4 Fix the Battery Pack

Material: Battery holder or battery pack.

Action: Place the battery pack near the center or rear side of the chassis to balance the car weight.

Check: Battery wires should not touch wheels or motor shafts.

Step 5 Place the IR Sensor

Material: One IR sensor module for obstacle avoiding or two sensors for better following.

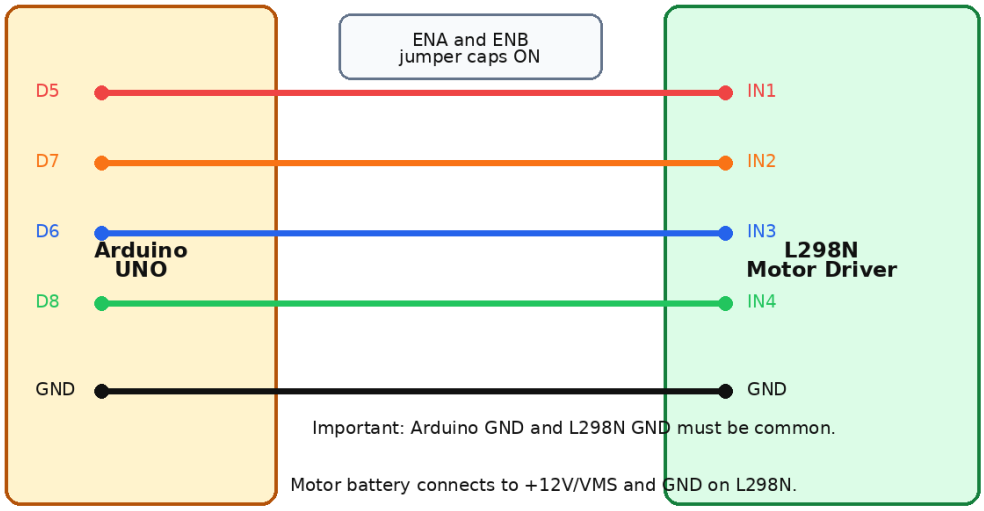
Action: Fix the IR sensor at the front of the chassis. The sensor LEDs should face forward toward the obstacle.

Check: Sensor should not be blocked by the chassis or wires.

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Wiring Manual

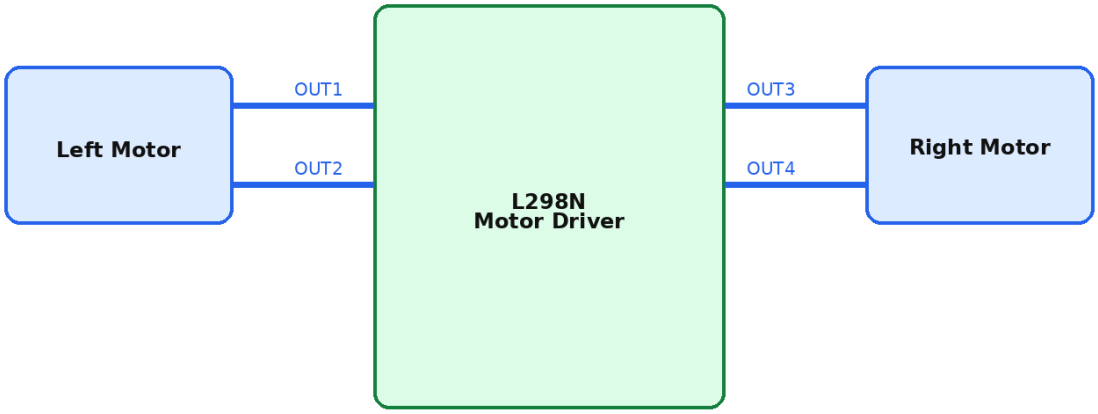
Arduino UNO to L298N Wiring



Arduino Uno to L298N wiring. Keep ENA and ENB jumper caps ON when speed pins are not connected.

Arduino Uno Pin	L298N Pin	Purpose
D5	IN1	Left motor direction input 1
D7	IN2	Left motor direction input 2
D6	IN3	Right motor direction input 1
D8	IN4	Right motor direction input 2
GND	GND	Common ground

Motor Output Connections



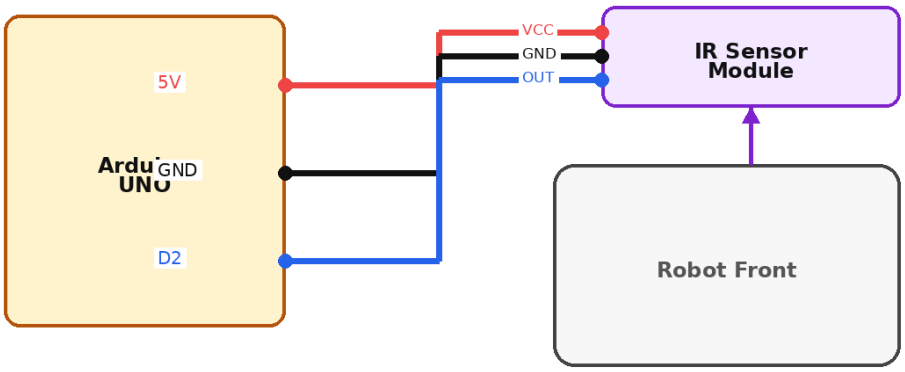
If the robot moves backward during FORWARD, swap any one motor's two output wires.

If LEFT and RIGHT are reversed, swap left and right motor sides in code or wiring.

L298N motor output wiring for left and right motors.

L298N Output	Connect to	Note
OUT1 and OUT2	Left motor two wires	Swap these two wires if left motor rotates opposite.
OUT3 and OUT4	Right motor two wires	Swap these two wires if right motor rotates opposite.
+12V or VMS	Motor battery positive	Use the motor battery positive wire.
GND	Motor battery negative and Arduino GND	Ground must be common.

IR Sensor Placement and Wiring



For obstacle avoiding, one front IR sensor is enough.

For better obstacle following, use two front sensors: Left IR on D2 and Right IR on D3.

IR sensor placement and wiring for obstacle avoiding and obstacle following projects.

IR Sensor Pin	Arduino Uno Connection	Purpose
VCC	5V	Power for sensor
GND	GND	Ground
OUT	D2	Sensor output signal

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Battery Connection

Connect motor battery positive to L298N +12V or VMS terminal. Connect motor battery negative to L298N GND. Connect Arduino GND to the same L298N GND terminal. Power Arduino through USB or a safe regulated supply.

Code Upload Steps

1. Open Arduino IDE.
2. Connect Arduino Uno to laptop using USB cable.
3. Select Tools > Board > Arduino Uno.
4. Select the correct COM port from Tools > Port.
5. Paste the code below and click Upload.
6. After upload, place the robot on floor and switch on motor battery.

Obstacle Avoiding Code

```
#define IR_SENSOR 2
#define IN1 5
#define IN2 7
#define IN3 6
#define IN4 8

void setup() {
  pinMode(IR_SENSOR, INPUT);
  pinMode(IN1, OUTPUT);
  pinMode(IN2, OUTPUT);
  pinMode(IN3, OUTPUT);
  pinMode(IN4, OUTPUT);
  stopRobot();
}

void loop() {
  int sensorValue = digitalRead(IR_SENSOR);

  if (sensorValue == LOW) {
    stopRobot();
    delay(200);
    backward();
    delay(400);
    right();
    delay(500);
  } else {
    forward();
  }
}

void forward() {
  digitalWrite(IN1, HIGH); digitalWrite(IN2, LOW);
  digitalWrite(IN3, HIGH); digitalWrite(IN4, LOW);
}

void backward() {
  digitalWrite(IN1, LOW); digitalWrite(IN2, HIGH);
  digitalWrite(IN3, LOW); digitalWrite(IN4, HIGH);
}

void right() {
  digitalWrite(IN1, HIGH); digitalWrite(IN2, LOW);
  digitalWrite(IN3, LOW); digitalWrite(IN4, HIGH);
}

void stopRobot() {
  digitalWrite(IN1, LOW); digitalWrite(IN2, LOW);
  digitalWrite(IN3, LOW); digitalWrite(IN4, LOW);
```

```
}
```

Obstacle Following Code

```
#define IR_SENSOR 2
#define IN1 5
#define IN2 7
#define IN3 6
#define IN4 8

void setup() {
  pinMode(IR_SENSOR, INPUT);
  pinMode(IN1, OUTPUT);
  pinMode(IN2, OUTPUT);
  pinMode(IN3, OUTPUT);
  pinMode(IN4, OUTPUT);
  stopRobot();
}

void loop() {
  int sensorValue = digitalRead(IR_SENSOR);

  if (sensorValue == LOW) {
    forward();
  } else {
    stopRobot();
  }
}

void forward() {
  digitalWrite(IN1, HIGH); digitalWrite(IN2, LOW);
  digitalWrite(IN3, HIGH); digitalWrite(IN4, LOW);
}

void stopRobot() {
  digitalWrite(IN1, LOW); digitalWrite(IN2, LOW);
  digitalWrite(IN3, LOW); digitalWrite(IN4, LOW);
}
```

Testing Procedure

Test	How to test	Correct result
Motor test	Lift the car and upload simple forward command.	Both wheels rotate forward.
Direction test	Run forward, backward, left, and right movement.	Robot follows the command correctly.
IR sensor test	Place hand or object in front of sensor.	Sensor LED changes and robot reacts.
Obstacle avoiding	Place an object in front of moving robot.	Robot stops, moves back, and turns.
Obstacle following	Move an object slowly in front of sensor.	Robot moves only when object is detected.

Troubleshooting

Problem	Possible reason	Solution
Motor not moving	No motor battery, loose motor wire, ENA ENB jumper missing	Check battery, OUT wires, and jumper caps.
Arduino resets	Motor battery load causing voltage drop	Power Arduino separately through USB or regulated supply.
Robot moves backward	Motor wires are reversed	Swap one motor output pair or change HIGH LOW logic in code.
Left and right reversed	Motor sides are swapped	Swap left and right motor wiring or code pin mapping.
IR sensor not detecting	Sensor distance knob not adjusted or light interference	Adjust potentiometer and test away from strong sunlight.
L298N heating	Wrong wiring or motor current too high	Stop power and recheck motor terminals and battery voltage.

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Trainer Notes

- For preschool workshops, demonstrate the robot movement first and then explain sensor action in simple words.
- Use the phrase: sensor is the eye, Arduino is the brain, motor driver is the muscle, and motors are the legs.
- Keep wires short and tied neatly so students can clearly see each component.

Learning Outcome

- Students understand the basic structure of a robot car.
- Students identify Arduino Uno, L298N motor driver, IR sensor, motors, and battery.
- Students learn how sensor input can control robot movement.
- Students observe obstacle avoiding and obstacle following logic in action.